



Instructor: Date:

Learner: Regional Branch:

Program: Center:

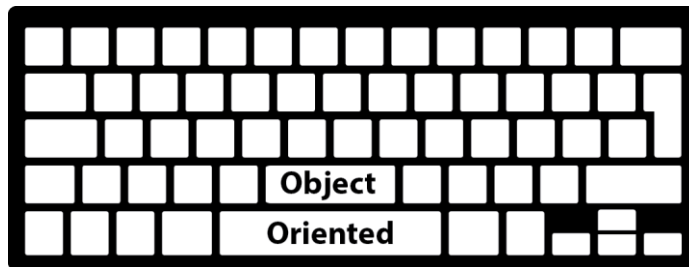
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Let's Start!

Let's learn about composite structure diagrams!



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Give detailed information related to UML and common related topics.

Ask your partner:

- Are you familiar with the changes between UML 1.x and 2.x? Are they relevant? Explain.
- How can you avoid confusions between the different kinds of diagrams?

Can Do Statements:

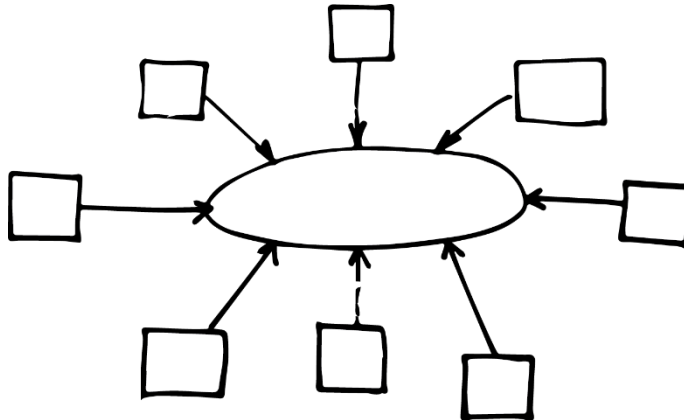
- I can explain how the composite structure diagram works.
- I can give details about the composite structure diagram.
- I can identify key features related to composite structure diagrams.



Let's Explore!

Two coworkers need to decide what model to use for their current* project.

*"Current" means the most recent.



Instructions: Read the text below.

John: Today I'm going to tell you about Composite structure diagrams. They're very useful when you've got to control run-time instances that collaborate by using communication links. Before we start talking about it, I will tell you that you haven't got to keep your questions to yourselves, you can ask any questions if you need to. So, let's get started! This kind of diagram works by enabling collaboration, which shows cooperation between different instances...

Sally: Excuse me, sir. Are these similar to collaboration diagrams?

John: They share some elements, but Collaboration diagrams are what we call Communication diagrams in UML 2. In fact, CS diagrams can show boxes containing a classifier which shows steps to follow to accomplish something. For example, the different stages to enroll in a course, such as the requirements, selection tests, form completion, etc. Any questions?

Sally: No, I haven't got any. However, what you're describing sounds very similar to the structure of an object role model diagram.

John: You're right Sally! But you haven't got to confuse them, it's important to note that this similarity mostly relates to their notation. Additionally, these two kinds of diagrams deal with issues and structures differently. For example, when you use ORM diagrams they will focus more on relationships between entities. On the other hand, CS diagrams are centered towards collaborations. Any other comments?

Sally: No. I guess I'll ask you for more guidance before I start using this diagram.

John: Great! Another element to take into account is the fact that editors prefer to use other diagrams to explore collaboration among elements. They say that CS diagrams haven't got a detailed notation. They say that the most appropriate kind of diagram to use in this case is sequence diagrams, especially because it's more widespread among developers. However, after you use this diagram you'll tell me whether you liked it or not.

**Let's Practice!**

John explains some concepts of CS diagrams.

Instructions: Use the words from the box to correctly complete the sentences.

Word Bank

classify / run-time / includes / Before / internal,

1. Before we start working let me tell you that we will use composite structure diagrams to control _____ instances.
2. After we finish, you will see how CS diagrams _____ the steps to accomplish something.
3. _____ that, A CS diagram will show you instances collaborating over communication links.
4. After we finish the lesson, the teacher will tell us that the composite structure diagram shows the _____ structure of a class.
5. After we are done programming, we will see that a CS diagram _____ a set of interconnected elements.

**Let's Practice!**

It's important to understand some terms in computer science.

Instructions: Organize the mystery sentences. Read the sentences aloud. Check your pronunciation with your instructor.

1. hasn't / An / to / editor / got / edit / file. / forbid / or / modify / you / to / a

2. system. / conditions / the / of / simulate / has / Emulation / another / to / got

3. be / prohibit / Entropy / got / to / to / encoded. / hasn't / data

4. system. / / to / another / imitate / Emulators / got / have





Extension Activity

You are going to make your own composite structure diagram.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about a process at your school or university.
- Check the parts and roles involved in the process
- Write at least 3 parts and 3 roles.
- Draw the process in a diagram using boxes and describe how they will collaborate and the steps to achieve the goal.
- Share your drawing with you classmates.

Self-Assessment

Yes

No

- I look for information about a process in my school or university.
- I draw a diagram related to a process.
- I share my findings with my partners.

☐	☐
☐	☐
☐	☐



Target Words

- **run-time (noun):** that happens during the execution of a program
e.g. They're very useful when you've got to control run-time instances that collaborate by using communication links.
- **cooperation (noun):** the act of working together towards a common goal
e.g. This kind of diagram works by enabling collaboration, which shows cooperation between different instances...
- **accomplish (verb):** to finish something successfully
e.g. CS diagrams can show boxes containing a classifier which shows steps to follow to accomplish something.
- **enroll (verb):** to enlist, to register
e.g. For example, the different stages to enroll in a course, such as the requirements, selection tests, form completion, etc.
- **notation (noun):** system of symbols that you use in specific fields
e.g. You're right Sally! But you haven't got to confuse them, it's important to note that this similarity mostly relates to their notation.
- **guidance (noun):** advice or counseling teachers or tutors provide to students
e.g. advice or counseling teachers or tutors provide to students



Learning Points

Future Time Clauses (Before and After)

You use the future tense to talk about distant future or things that you haven't planned.

Future time clauses are dependent clauses that you can link to an independent clause; the *independent clause* is the one you write using *future simple*, and you connect it to the independent clause using *Before* and *After*.

You can use a simple present sentence or you can use a verb in *Ing form* after "Before" and "After".

Examples:

*I'll start using that diagram **after** we reach an agreement.*

*I'll start using that diagram **after** reaching an agreement.*

*We won't discard the design **before** we finish the project.*

*We won't discard the design **before** finishing the project.*

Examples

- *Before we start talking about it, I will tell you that you haven't got to keep your questions to yourselves, you can ask any questions if you need to.*
- *No. I guess I'll ask you for more guidance before I start using this diagram.*





Learning Points

Have Got (Negative Sentences)

You use Have Got to talk about obligations and necessity.

In order to use **Have got**, you need to take into account that the verb Have changes depending on the subject it follows:

Pronoun	Have
I	<i>have</i>
You	<i>have</i>
He	<i>has</i>
She	<i>has</i>
It	<i>has</i>
We	<i>have</i>
They	<i>have</i>

You can use Have Got in negative sentences to show that there is no obligation to do something, or also to show that you don't have possession of a noun.

Examples:

*I **haven't got** to use this type of diagram because I have got to use the composite structure diagram.*

*She **hasn't got** to know all the types of diagrams.*

*He **hasn't got** any computers to work with.*

Examples

- You're right Sally! But you haven't got to confuse them.*
- Before we start talking about it, I will tell you that you haven't got to keep your questions to yourselves, you can ask any questions if you need to.*



Learning Points

List of Programming and Computer Science Terms

There are many terms that computer science has adopted to refer to the different activities or processes that surround it. You need to know them to be able to communicate with your colleagues and even understand manuals or instructions.

Editor	An editor can be a computer program that allows you to edit or modify a file. It can also refer to the person who makes those modifications.
Emacs	It refers to a group of text editors that contain over 2000 commands which is commonly used in macros to make work automatic.
Emulation	It is when a system simulates the conditions and behaviors of another one. It also includes the use of virtual machines.
Emulator	A piece of hardware or software that imitates or simulates the behaviors of another one. In this case, the system that wants to imitate the other is called the host, and the one that is simulated is called the guest.
Entropy	A condition of randomness in computing that allows for data to be encoded. It is a key component in security and performance.

Example

- *Another element to take into account is the fact that editors prefer to use other diagrams to explore collaboration among elements.*

